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Method of performing full firmware update procedure on embedded electronic device and related embedded electronic device

Abstract

When performing full firmware update procedure on an electronic device, a full firmware update file is divided into multiple pieces of update data and then arranged into a first and a second sub update files, a header file associated with the sizes of each sub update file is created, and the header file and the two sub update files are merged into a full FOTA file before being uploaded to the electronic device. The electronic device extracts each sub update file based on the header file and stores the extracted first and second sub update file respectively into a first and a second storage units. After performing the firmware update steps associated with the first sub update file, the second sub update file stored in the second storage unit is copied to the first storage unit before performing the firmware update steps associated with the second sub update file.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/458,166, filed on Apr. 10, 2023. The content of the application

is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention is related to a method of performing full firmware update procedure on an embedded electronic device and a related embedded electronic device, and more particularly, to a method of performing full firmware update procedure using FOTA technology on an embedded electronic device with limited built-in memory space and a related embedded electronic device.

2. Description of the Prior Art

(2) An embedded electronic device typically includes hardware, software, and firmware. Hardware generally refers to all the physical components required for a system to function, such as motherboards, graphics cards, CPUs, fans, webcams, and power supplies, each of which may be connected to each other through circuit boards. Software generally refers to non-physical products created in programming languages, such as applications (APPs), web pages, and various operating systems which are used to enable hardware to provide users with various services. Firmware refers to the software embedded in a hardware device, and may control the hardware device through the programming language. Firmware is commonly installed in a basic input/output system chip, an optical drive, a burner, an Internet protocol gateway/router, a wireless access point (WAP) and a modem on the motherboard. A user can remotely control and operate the settings and functions of the firmware through a designated interface, monitor the current operational status of the hardware and set/upgrade/adjust the hardware.

(3) Firmware may be viewed as a kit which contains all the official proprietary software and is usually stored in the flash memory of the embedded electronic device. The user may perform a firmware update procedure through a specific device and a specific method. The main purpose of firmware updates includes correcting errors in the source codes, adding or improving execution functions, adjusting system security, and improving firmware efficiency.

(4) In a first prior art method of updating the firmware, the embedded electronic device is connected to the computer via a universal serial bus (USB) port. Next, the firmware update procedure may be performed off-line by downloading the latest version of the firmware file in the official FTF format using the FlashTool flashing software. This prior method can transmit large amount of firmware files for a full firmware update procedure. However, such wired data transmission is not feasible or extremely inconvenient in certain application fields.

(5) In a second prior art method, a firmware over-the-air (FOTA) technology is used to automatically update the operating system of the embedded electronic device. By connecting to a server, the embedded electronic device may download the latest version of the firmware file and update the firmware accordingly without connecting to the computer host. However, if an error occurs in the root file system (rootfs) or a modem of the embedded electronic device, the firmware update procedure may not be able to be performed on-line. In addition, the storage space of the built-in flash memory of a mobile electronic device is usually insufficient to support the file size required for a full firmware update procedure using FOTA technology. Therefore, an electronic device with limited built-in memory space can only use the first prior art method when performing a full firmware update procedure, which is inconvenient for the users.

(6) Therefore, there is a need for a method of performing a full firmware update procedure on an electronic device with limited built-in memory space using FOTA technology.

SUMMARY OF THE INVENTION

(7) The present invention provides a method of performing a full firmware update procedure on an embedded electronic device. The method includes dividing a full firmware update file into a plurality pieces of update data; arranging the plurality pieces of update data into a first sub update file and a second sub update file; creating a header file associated with a size of each sub update file; merging the header file, the first sub update file and the second sub update file sequentially

into a full FOTA file; uploading the full FOTA file to the embedded electronic device; extracting the first sub update file from the full FOTA file based on the header file and storing the extracted first sub update file in a first storage unit of the embedded electronic device; freeing up memory space of a second storage unit in a sub-system module of the embedded electronic device; extracting the second sub update file from the full FOTA file based on the header file and storing the extracted second sub update file in the second storage unit; performing one or multiple firmware update steps associated with the first sub update file stored in the first storage unit at a first time point; copying the second sub update file stored in the second storage unit to the first storage unit at a second time point which occurs after the first time point; and performing one or multiple firmware update steps associated with the second sub update file stored in the first storage unit at a third time point which occurs after the second time point. The first storage unit provides a first available memory space during an operation of the embedded electronic device. The second storage unit provides a second available memory space during the operation of the embedded electronic device. A size of the full FOTA file is larger than a size of the first available memory space. The size of the full FOTA file is not larger than a total size of the first available memory space and the second available memory space.

(8) The present invention also provides an embedded electronic device which performs a full firmware update procedure. The embedded electronic device includes a first storage unit configured to provide a first available memory space during an operation of the embedded electronic device, a sub-system module having a built-in second storage unit configured to provide a second available memory space during the operation of the embedded electronic device, and a processing module. The processing module configured to receive a full FOTA file associated with a full firmware update file and including a header file, a first sub update file and a second sub update file; extract the first sub update file from the full FOTA file based on the header file and store the extracted first sub update file in the first storage unit; free up memory space of the second storage unit in the sub-system module; extract the second sub update file from the full FOTA file based on the header file and store the extracted second sub update file in the second storage unit; perform one or multiple firmware update steps associated with the first sub update file stored in the first storage unit at a first time point; copy the second sub update file stored in the second storage unit to the first storage unit at a second time point which occurs after the first time point; and perform one or multiple firmware update steps associated with the second sub update file stored in the first storage unit at a third time point which occurs after the second time point. The full firmware update file is divided into a plurality pieces of update data. The plurality pieces of update data are arranged into the first sub update file and the second sub update file. The header file is associated with a size of each sub update file. A size of the full FOTA file is larger than a size of the first available memory space. The size of the full FOTA file is not larger than a total size of the first available memory space and the second available memory space.

(9) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a functional diagram illustrating an embedded electronic device according to an embodiment of the present invention.

(2) FIG. 2 is a flowchart illustrating a method of performing a full firmware update procedure on an embedded electronic device using FOTA technology according to an embodiment of the present invention.

(3) FIG. 3 is a diagram illustrating the data structure in the embedded electronic device after executing the file-dividing step of the full firmware update procedure according to an embodiment of the present invention.

(4) FIG. 4 is a diagram illustrating the full FOTA file created by the embedded electronic device when performing the full firmware update procedure according to an embodiment of the present invention.

(5) FIGS. 5-9 are diagrams illustrating the data structures in the embedded electronic device at each stage of the full firmware update procedure using FOTA technology according to an embodiment of the present invention.

(6) FIG. 10 is a diagram illustrating the data structure in the embedded electronic device after performing the full firmware update procedure according to an embodiment of the present invention.

DETAILED DESCRIPTION

(7) FIG. 1 is a functional diagram illustrating an embedded electronic device **100** according to an embodiment of the present invention. The embedded electronic device **100** includes at least a processing module **110**, a sub-system module **120**, a first storage unit **131** and a user interface **140**.

(8) In an embodiment of the present invention, the embedded electronic device **100** may be an Internet of Things (IoT) device, a mobile Internet device (MID) or a smart phone designed for industrial, medical storage or automobile application. However, the type and application of the embedded electronic device **100** do not limit the scope of the present invention.

(9) In an embodiment of the present invention, the sub-system module **120** includes at least a second storage unit **132** for storing sub-system firmware which provides functions required by the application of the embedded electronic device **100**. For example, when the embedded electronic device **100** is an Internet Service Provider (ISP), the sub-system module **120** may be a modem module. However, the types of the embedded electronic device **100** and its built-in sub-system module **120** do not limit the scope of the present invention.

(10) In an embodiment of the present invention, the first storage unit **131** may be a built-in memory of the embedded electronic device **100**, such as an embedded multimedia card (eMMC), a universal flash storage (UFS) or a non-volatile memory (NVM). However, the type of the first storage unit **131** does not limit the scope of the present invention.

(11) In an embodiment of the present invention, the second storage unit **132** may be a built-in memory of the sub-system module **120** and may include one or multiple types of memory devices. For example, the second storage unit **132** may include flash memory, random access memory (RAM) and/or read-only memory (ROM). However, the type of the second storage unit **132** does not limit the scope of the present invention.

(12) In an embodiment of the present invention, the embedded electronic device **100** is a product designed for ultra-thin and light-weight applications, wherein its built-in first storage unit **131** only provides limited available memory space. More specifically, during the operation of the embedded electronic device **100**, the first storage unit **131** is unable to provide sufficient available memory space for performing a full firmware update procedure.

(13) In an embodiment of the present invention, during the operation of the embedded electronic device **100**, the available memory space provided by the first storage unit **131** is smaller than the minimum memory space required for performing the full firmware update procedure, but the total available memory space provided by the first storage unit **131** and the second storage unit **132** is larger than the minimum memory space required for performing the full firmware update procedure. Therefore, when the embedded electronic device **100** is requested to perform the full firmware update procedure, the present invention may further use the second storage unit **132** of the sub-system module **120** to provide extra memory space for supporting the full firmware update procedure.

(14) FIG. 2 is a flowchart illustrating a method of performing a full firmware update procedure on

the embedded electronic device **100** using FOTA technology according to an embodiment of the present invention. The flowchart depicted in FIG. 2 includes the following steps:

(15) Step **200**: divide a full firmware update file into a plurality pieces of update data.

(16) Step **210**: arrange the plurality pieces of update data into a first sub update file and a second sub update file.

(17) Step **220**: create a header file associated with the firmware version, a checksum and the size of each sub update file.

(18) Step **230**: merge the header file, the first sub update file and the second sub update file sequentially into a full FOTA file.

(19) Step **240**: upload the full FOTA file to the embedded electronic device **100**.

(20) Step **250**: determine whether the full FOTA file can pass an accuracy verification; if yes, execute step **260**; if no, execute step **320**.

(21) Step **260**: extract the first sub update file from the full FOTA file based on the header file and store the extracted first sub update file in the first storage unit **131**.

(22) Step **270**: free up the memory space of the second storage unit **132** of the sub-system module **120**.

(23) Step **280**: extract the second sub update file from the full FOTA file based on the header file and store the extracted second sub update file in the second storage unit **132**.

(24) Step **290**: determine whether the extracted first sub update file and the extracted second sub update file are complete; if yes, execute step **300**; if no, execute step **320**.

(25) Step **300**: restart the system and perform one or multiple firmware update steps associated with the first sub update file stored in the first storage unit **131**.

(26) Step **310**: restart the system, copy the second sub update file stored in the second storage unit **132** to the first storage unit **131**, and perform one or multiple firmware update steps associated with the second sub update file stored in the first storage unit **131**.

(27) Step **320**: send a notification of firmware update failure and exit the full firmware update procedure.

(28) In step **200**, the full firmware update file is divided into the plurality pieces of update data. Each piece of update data may be any one among a kernel update data, a root file system (rootfs) update data, an original equipment manufacturer (OEM) update data and a sub-system firmware update data. In an embodiment, the full firmware update file may be divided into the plurality pieces of update data using any file splitter tool or software in step **200**.

(29) In step **210**, the plurality pieces of update data are arranged into the first sub update file and the second sub update file. For illustrative purpose, it is assumed that the full firmware update file is divided into $(M+N)$ pieces of update data, wherein the first sub update file includes N pieces of update data, the second sub update file includes M pieces of update data, and M and N are positive integers. The content of each piece of update data in the first sub update file is different from the content of each piece of update data in the second sub update file. As previously stated, the minimum memory space required for performing the full firmware update procedure is larger than the size of the available memory space provided by the first storage unit **131**, but is not larger than the total size of the available memory space provided by the first storage unit **131** and the second storage unit **132**. In an embodiment, the sizes of the first sub update file and the second sub update file (i.e., the values of M and N) may be determined according to the ratio of the size of the first available memory space provided by the first storage unit **131** to the size of the second available memory space provided by the second storage unit **132**.

(30) In an embodiment of the present invention, the second sub update file includes the sub-system firmware update data of the full firmware update file, wherein the one or multiple firmware update steps associated with the first sub update file are performed earlier than the one or multiple firmware update steps associated with the second sub update file.

(31) FIG. 3 is a diagram illustrating the data structure in the embedded electronic device **100** after

executing step **210** of the full firmware update procedure according to an embodiment of the present invention. For illustrative purpose, it is assumed that the full firmware update file is divided into 4 pieces of update data: the rootfs update data, the OEM update data, the kernel update data, and the sub-system firmware update data, wherein the total size of the full firmware update file is about 144 MB. Assuming that the size of the first available memory space provided by the first storage unit **131** is similar to the size of the second available memory space provided by the second storage unit **132**, the first sub update file may include the rootfs update data and the OEM update data (with a total size of about 73 MB), and the second sub update file may include the kernel update data and the sub-system firmware update data (with a total size of about 71 MB) after executing step **210**, as depicted in FIG. 3.

(32) In the embodiment of the present invention, the content of each piece of update data in the first sub update file is different from the content of each piece of update data in the second sub update file, and the second sub update file at least includes the sub-system firmware update data.

(33) In step **220**, the header file associated with the firmware version, the checksum and the size of each sub update file is created. In step **230**, the header file, the first sub update file and the second sub update file are merged sequentially into the full FOTA file.

(34) FIG. 4 is a diagram illustrating the full FOTA file created by the embedded electronic device **100** when performing the full firmware update procedure according to an embodiment of the present invention. In an embodiment, the checksum may be cyclic redundancy check (CRC) for detecting accidental errors to digital data after data transmission or storage. In another embodiment, the checksum may be any cryptographic hash function containing fixed string of digits which are generated based on a network data package or a computer file. In an embodiment, Size **1** represents the file size of the first sub update file and its value may be associated with the number of bits contained in the first sub update file. Size **2** represents the file size of the second sub update file and its value may be associated with the number of bits contained in the second sub update file. However, the format of the header file and the type of the checksum do not limit the scope of the present invention.

(35) In step **240**, the full FOTA file may be uploaded to the embedded electronic device **100** via the user interface **140**. In step **250**, it is then determined whether the full FOTA file can pass the accuracy verification. In an embodiment, the processing unit **110** of the embedded electronic device **100** may examine the content of the header file in the full FOTA file and verify the correctness of the checksum. When determining that the full FOTA file cannot pass the accuracy verification, it indicates that data loss or damage may have happened during the process of uploading the full FOTA file to the embedded electronic device **100**. Under such circumstance, step **320** is then executed for sending the notification of firmware update failure and exiting the full firmware update procedure.

(36) When it is determined in step **250** that the full FOTA file can pass the accuracy verification, steps **260** and **280** are then executed for respectively extracting the first sub update file and the second sub update file from the full FOTA file based on the header file. As depicted in the embodiment of FIG. 4, the processing unit **110** of the embedded electronic device **100** may acquire the file size of the first sub update file (i.e., Size **1**) and the file size of the second sub update file (i.e., Size **2**) based on the header file of the full FOTA file, and the extract corresponding contents of the first sub update file and the second sub update file from the full FOTA file based on Size **1** and Size **2**. For example, when the value of Size **1** is equal to m and the value of Size **2** is equal to n (m and n are positive integers), the processing unit **110** may extract the first bit to the m .sup.th bit of data located after the header file as the first sub update file, and may extract the $(m+1)$.sup.th bit to the n .sup.th bit of data located after the header file as the second sub update file.

(37) FIGS. 5-9 are diagram illustrating the data structures in the embedded electronic device **100** at each stage of the full firmware update procedure using FOTA technology according to an embodiment of the present invention. For illustrative purpose, it is assumed that current rootfs data,

current OEM data, current kernel data, and current sub-system firmware data is stored in the embedded electronic device **100** before performing the full firmware update procedure.

(38) In step **260**, the first sub update file extracted from the full FOTA file is stored in the first storage unit **131** of the embedded electronic device **100**, as depicted in FIG. 5.

(39) In step **270**, the memory space of the second storage unit **132** in the sub-system module **120** is freed up. In step **280**, the second sub update file extracted from the full FOTA file is stored in the second storage unit **132** of the sub-system module **120**, as depicted in FIG. 6.

(40) In step **290**, the processing unit **110** is configured to determine whether the extracted first sub update file and the extracted second sub update file are complete. When determining that the extracted first sub update file and the extracted second sub update file are not complete, it indicates that data loss or damage may have happened during the process of extracting and/or storing each sub update file. Under such circumstance, step **320** is then executed for sending the notification of firmware update failure and exiting the full firmware update procedure.

(41) When it is determined in step **290** that the extracted first sub update file and the extracted second sub update file are complete, step **300** is executed for restarting the system and performing the one or multiple firmware update steps associated with the first sub update file stored in the first storage unit **131**. More specifically, after restarting the system in step **300**, the processing unit **110** is configured to update the current rootfs data according to the rootfs update data in the first sub update file and update the current OEM data according to the OEM update data in the first sub update file, as depicted in FIG. 7.

(42) After performing the one or multiple firmware update steps associated with the first sub update file, step **310** is executed for restarting the system again. Next, the second sub update file stored in the second storage unit **132** is copied to the first storage unit **131**, as depicted in FIG. 8.

(43) Next, the one or multiple firmware update steps associated with the second sub update file stored in the first storage unit **131** may be performed. More specifically, after copying the second sub update file stored in the second storage unit **132** to the first storage unit **131**, the processing unit **110** is configured to update the current kernel data according to the kernel update data in the second sub update file and update the current sub-system module **120** according to the sub-system update data in the second sub update file, as depicted in FIG. 9.

(44) In an embodiment, after restarting the system in steps **300** and **310**, the embedded electronic device **100** is configured to perform an original equipment manufacturer secondary boot loader (OEMSBL) for performing the one or multiple firmware update steps associated with the first sub update file and the second sub update file. However, the method of performing the one or multiple firmware update steps associated with the first sub update file and the second sub update file adopted by the embedded electronic device **100** does not limit the scope of the present invention.

(45) FIG. 10 is a diagram illustrating the data structure in the embedded electronic device **100** after performing the full firmware update procedure using FOTA technology according to an embodiment of the present invention. After completing the full firmware update procedure, the updated rootfs data, the updated OEM data, the updated kernel data, and the updated sub-system firmware is stored in the embedded electronic device **100**.

(46) In conclusion, for an embedded electronic device with limited built-in memory space, the present invention performs the full firmware update procedure using a multi-stage FOTA technology. The firmware update steps associated with the sub-system firmware update data are performed at the last stage, so that the second storage unit **132** of the sub-system module **120** may provide extra memory space for supporting the full firmware update procedure before updating the current sub-system.

(47) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A method of performing a full firmware update procedure on an embedded electronic device, comprising: dividing a full firmware update file into a plurality pieces of update data; arranging the plurality pieces of update data into a first sub update file and a second sub update file; creating a header file associated with a size of each sub update file; merging the header file, the first sub update file and the second sub update file sequentially into a full firmware over-the-air (FOTA) file; uploading the full FOTA file to the embedded electronic device; extracting the first sub update file from the full FOTA file based on the header file and storing the extracted first sub update file in a first storage unit of the embedded electronic device; freeing up memory space of a second storage unit in a sub-system module of the embedded electronic device; extracting the second sub update file from the full FOTA file based on the header file and storing the extracted second sub update file in the second storage unit; performing one or multiple firmware update steps associated with the first sub update file stored in the first storage unit at a first time point; copying the second sub update file stored in the second storage unit to the first storage unit at a second time point which occurs after the first time point; and performing one or multiple firmware update steps associated with the second sub update file stored in the first storage unit at a third time point which occurs after the second time point, wherein: the first storage unit provides a first available memory space during an operation of the embedded electronic device; the second storage unit provides a second available memory space during the operation of the embedded electronic device; a size of the full FOTA file is larger than a size of the first available memory space; and the size of the full FOTA file is not larger than a total size of the first available memory space and the second available memory space.
2. The method of claim 1, wherein: the full firmware update file is divided into $(M+N)$ pieces of update data; the first sub update file includes N pieces of update data; the second sub update file includes M pieces of update data; M and N are positive integers; the M pieces of update data at least includes a sub-system firmware update data; and a content of each piece of update data in the first sub update file is different from a content of each piece of update data in the second sub update file.
3. The method of claim 2, wherein: the N pieces of update data in the first sub update file includes a kernel update data, a root file system update data or an original equipment manufacturer (OEM) update data; and the M pieces of update data in the second sub update file further includes any of the kernel update data, the root file system update data and the OEM update data which is not included in the N pieces of update data of the first sub update file.
4. The method of claim 1, further comprising: determining a value of M and a value of N according to a ratio of the size of the first available memory space to the size of the second available memory space.
5. The method of claim 1, further comprising: creating the header file associated with the size of each sub update file and a checksum; determining whether the full FOTA file can pass an accuracy verification after uploading the full FOTA file to the embedded electronic device; and extracting the first sub update file from the full FOTA file and extracting the second sub update file from the full FOTA file according to the header file after determining that the full FOTA file can pass the accuracy verification.
6. The method of claim 5, further comprising: sending a notification of firmware update failure and exiting the full firmware update procedure when determining that the full FOTA file cannot pass the accuracy verification after being uploaded to the embedded electronic device.
7. The method of claim 1, further comprising: determining whether the first sub update file is complete after storing the extracted first sub update file in the first storage unit; and after determining that the first sub update file is complete, restarting the embedded electronic device at

the first time point for performing the one or multiple firmware update steps associated with the first sub update file stored in the first storage unit.

8. The method of claim 7, further comprising: sending a notification of firmware update failure and exiting the full firmware update procedure when determining that the first sub update file is not complete after being stored in the first storage unit.

9. The method of claim 1, further comprising: determining whether the second sub update file is complete after copying the extracted second sub update file to the first storage unit; and after determining that the second sub update file is complete, restarting the embedded electronic device at the third time point for performing the one or multiple firmware update steps associated with the second sub update file stored in the first storage unit.

10. The method of claim 9, further comprising: sending a notification of firmware update failure and exiting the full firmware update procedure when determining that the second sub update file is not complete after being copied to the first storage unit.

11. An embedded electronic device which performs a full firmware update procedure, comprising: a first storage unit configured to provide a first available memory space during an operation of the embedded electronic device; a sub-system module having a built-in second storage unit configured to provide a second available memory space during the operation of the embedded electronic device; and a processing module configured to: receive a full firmware over-the-air (FOTA) file associated with a full firmware update file and including a header file, a first sub update file and a second sub update file; extract the first sub update file from the full FOTA file based on the header file and store the extracted first sub update file in the first storage unit; free up memory space of the second storage unit in the sub-system module; extract the second sub update file from the full FOTA file based on the header file and store the extracted second sub update file in the second storage unit; perform one or multiple firmware update steps associated with the first sub update file stored in the first storage unit at a first time point; copy the second sub update file stored in the second storage unit to the first storage unit at a second time point which occurs after the first time point; and perform one or multiple firmware update steps associated with the second sub update file stored in the first storage unit at a third time point which occurs after the second time point, wherein: the full firmware update file is divided into a plurality pieces of update data; the plurality pieces of update data are arranged into the first sub update file and the second sub update file; the header file is associated with a size of each sub update file; a size of the full FOTA file is larger than a size of the first available memory space; and the size of the full FOTA file is not larger than a total size of the first available memory space and the second available memory space.

12. The embedded electronic device of claim 11, wherein: the full firmware update file is divided into $(M+N)$ pieces of update data; the first sub update file includes N pieces of update data; the second sub update file includes M pieces of update data; M and N are positive integers; the M pieces of update data at least includes a sub-system firmware update data; and a content of each piece of update data in the first sub update file is different from a content of each piece of update data in the second sub update file.

13. The embedded electronic device of claim 12, wherein: the N pieces of update data in the first sub update file includes a kernel update data, a root file system update data or an original equipment manufacturer (OEM) update data; and the M pieces of update data in the second sub update file further includes any of the kernel update data, the root file system update data and the OEM update data which is not included in the N pieces of update data of the first sub update file.

14. The embedded electronic device of claim 11, wherein a value of M and a value of N are determined according to a ratio of the size of the first available memory space to the size of the second available memory space.

15. The embedded electronic device of claim 11, wherein: the header file is associated with the size of each sub update file and a checksum; and the processing module is further configured to: determine whether the full FOTA file can pass an accuracy verification after receiving the full

FOTA file; and extract the first sub update file from the full FOTA file and extracting the second sub update file from the full FOTA file according to the header file after determining that the full FOTA file can pass the accuracy verification.

16. The embedded electronic device of claim 11, wherein the processing module is further configured to send a notification of firmware update failure and exit the full firmware update procedure when determining that the full FOTA file cannot pass the accuracy verification after receiving the full FOTA file.

17. The embedded electronic device of claim 11, wherein the processing module is further configured to: determine whether the first sub update file is complete after storing the extracted first sub update file in the first storage unit; and after determining that the first sub update file is complete, restart the embedded electronic device at the first time point for performing the one or multiple firmware update steps associated with the first sub update file stored in the first storage unit.

18. The embedded electronic device of claim 17, wherein the processing module is further configured to send a notification of firmware update failure and exit the full firmware update procedure when determining that the first sub update file is not complete after being stored in the first storage unit.

19. The embedded electronic device of claim 11, wherein the processing module is further configured to: determine whether the second sub update file is complete after copying the extracted second sub update file to the first storage unit; and after determining that the second sub update file is complete, restart the embedded electronic device at the third time point for performing the one or multiple firmware update steps associated with the second sub update file stored in the first storage unit.

20. The embedded electronic device of claim 19, wherein the processing module is further configured to send a notification of firmware update failure and exit the full firmware update procedure when determining that the second sub update file is not complete after being copied to the first storage unit.
